

Ricky Faure

Principal Software Engineer

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SUMMARY

I build scalable enterprise-grade platforms with cutting edge open source tooling. Experienced in Kubernetes hosted applications, Operator Development, CI/CD, Terraform, and Distributed Systems.

SKILLS

Core: Go Kubernetes Kubernetes Operators Kubebuilder / Operator SDK Kubernetes CRDs Terraform
custom Terraform Provider development GCP GKE Apache Kafka Elasticsearch Helm CI/CD GitHub Actions
Python distributed systems SRE infrastructure as code architecture design

Additional: Docker Prometheus GitOps bare-metal Kubernetes GCP IAM GCS VPC Artifact Registry (GAR)
GCP VolumeSnapshots certificate management Jenkins Java Spring Boot Groovy Spock Kibana
Rally API automation vulnerability management mentorship and technical teaching AI-assisted development
agentic development workflows Claude Code GitHub Copilot OpenAI Codex

EXPERIENCE

Optum, UnitedHealth Group

JAN 2020 — PRESENT

Principal Software Engineer, TLCP — Technology Development Program Associate (Jan 2020) → Software Engineer (Jan 2021) → Senior Software Engineer (Mar 2022) → Lead Software Engineer (Feb 2023) → Principal Software Engineer, TLCP (Feb 2024 - Present)

Empower hundreds of teams across the enterprise to build data-driven services, moving tens of petabytes of data with minimal architectural overhead, using Kubernetes Operators and CI/CD to automate deployment, management, and hosting of enterprise-grade distributed systems across on-premise and multi-cloud environments.

- Platform and team leadership: Lead a team of 7 engineers building Kafka-as-a-Service and Elasticsearch-as-a-Service platforms, achieving five nines reliability via Kubernetes Operators and robust infrastructure-as-code practices.
- Kubernetes Operator development: Build and maintain custom Kubernetes Operators (Go, Kubebuilder) for Kafka, Elasticsearch, Prometheus, Service Monitors, GCP VolumeSnapshots, certificate management, and bare-metal Kubernetes upgrade orchestration.
- Control plane architecture: Design and manage the architecture of the platform's Kubernetes resource management control plane, in which operators generate and own concrete Kubernetes resources for automated deployment and configuration of streaming infrastructure at production scale, managing 800+ clusters across thousands of nodes in multi-tenant on-premise and GCP environments.
- Self-service marketplace migration: Achieved a 52% increase in resource deployments by migrating from GitOps to a GUI-based management system within an enterprise marketplace, backed by a custom Terraform Provider and Kubernetes resource manager.
- Cloud cost optimization: Delivered \$2.5M in annual cost savings through cloud resource optimization, instance type migrations, and elimination of excessive log retention in GCP.
- Cloud scale and GKE orchestration: Expanded the platform to handle 20+ petabytes of data movement across on-premise and GCP (GKE, GAR, IAM, GCS, VPC) by extending Kubernetes Operators for cloud environments; own provisioning and orchestration of the platform's GKE environments through Terraform and infrastructure as code.
- SRE and production operations: Serve in on-call rotation across a fleet of 800+ clusters and thousands of nodes, delivering five nines reliability with zero customer data loss in production across the platform's history; respond to incidents, tune performance and utilization against SLAs, and stabilize high-throughput Kafka and Elasticsearch services.

- **Agentic development workspaces:** Design and build agent-driven development workspaces that standardize the team's engineering workflow into a consistent, deterministic process; encode the full breadth of the platform's distributed projects and their interdependencies into durable agent context, and author reusable agent skills that automate remediation and SRE support tasks.
- **Security automation:** Designed a vulnerability scanning workstream using reusable GitHub Actions, a custom Python Rally API library, and automated report generation, adopted org-wide.
- **Mentorship and teaching:** Mentor junior engineers in rotational programs; teach quarterly internal courses on Kubernetes Operators and Elasticsearch/Kibana.
- **Featured Project: Warpstream Cluster Provisioning Platform (Q4 2025 - Q1 2026)**
- **Architected the end-to-end design for Warpstream-based Kafka cluster provisioning:** the networking design, the operator, agent, and resource specifications, and the resource flows from the customer-facing resource manager down to the backend infrastructure.
- **Extended the platform's Kubernetes operators for net-new resource kinds and delivered a net-new Warpstream operator (Go) that generates all required Kubernetes resources and connects them into Cloud Storage buckets, Warpstream API registrations, and agent configs.**
- **Co-designed the Terraform cloud infrastructure for the Warpstream Clusters-as-a-Service deployment, including GCP Cloud Storage provisioning, VPC configuration, DNS, and IAM.**
- **Shipped to two of Optum's largest Apache Kafka on GCP customers as beta; Warpstream's diskless, Cloud Storage-based architecture eliminates local disk I/O from the stream processing data path, projected to reduce their annual Kafka infrastructure spend by approximately 80% .**
- **Featured Project: Kafka Cluster Provisioning Platform on GCP**
- **Build and operate the Kafka-as-a-Service provisioning platform on GCP: custom Kubernetes Operators (Go) own the full cluster lifecycle on GKE, integrated with GCS, IAM, VPC, and Artifact Registry through Terraform-managed cloud infrastructure.**
- **Expose provisioning as self-service through the enterprise marketplace, backed by the custom Terraform Provider and Kubernetes resource management control plane.**
- **Featured Project: Kafka Cluster Provisioning Platform on Bare-Metal**
- **Build and operate the on-premise Kafka-as-a-Service platform on bare-metal Kubernetes: operators automate cluster provisioning, configuration, certificate management, and rolling upgrades across multi-tenant environments.**
- **Keep streaming workloads available through node cycling via the custom bare-metal Kubernetes upgrade orchestration operator.**
- **Featured Project: Elasticsearch Provisioning Platform on GCP**
- **Extended the Elasticsearch-as-a-Service operators into GCP, provisioning clusters on GKE with Terraform-managed cloud infrastructure and operator-driven GCP VolumeSnapshot backup and restore.**
- **Featured Project: Elasticsearch Provisioning Platform on Bare-Metal**
- **Build and operate Elasticsearch-as-a-Service on bare-metal Kubernetes: operators automate cluster provisioning, configuration, certificate management, and Kibana access, with Prometheus and Service Monitor operators providing observability across the fleet.**

Technologies: Go, Kubebuilder, Kubernetes Operators, Helm, Terraform, GCP, GKE, GitHub Actions, Python, Kafka, Warpstream, Elasticsearch, Prometheus, Jenkins, Docker

Optum

JUN — AUG, 2017 — 2019

TDP Software Development Intern

- Returned for three consecutive summer internships (June - August, 2017, 2018, and 2019) with similar roles and responsibilities each year.
- Contributed business-critical features to a monolithic application, complete with feature implementations and automated unit and integration testing.

Technologies: Java, Spring Boot, Groovy Spock

PROJECTS

OSS Feed

JavaScript · GitHub Actions · RSS/Atom · GitHub Pages · YAML

Self-hosted weekly RSS digest that tracks releases from curated open source projects. A watchlist YAML feeds GitHub Actions to fetch releases, generate an XML feed, and publish via GitHub Pages, with Slack/Discord webhook notifications and state management to avoid duplicate alerts.

Rubik's Cube Genetic Algorithm Solver

Python · genetic algorithms · matrix math

Evolutionary algorithm that solves a fully simulated Rubik's Cube model, exploring genetic computation applied to combinatorial optimization.

SafetyNet

Swift · iOS · P2P networking · E2E encryption

iOS peer-to-peer messaging app requiring no Wi-Fi or cellular data. Uses device-to-device networking with end-to-end encryption for resilient off-grid communication.

Ray Tracing Engine

JavaScript · graphics pipeline · 3D rendering · Blinn-Phong

Browser-based ray tracer in vanilla JavaScript. Renders spheres and triangles with Blinn-Phong lighting, reflections, and a free-moving camera, no GPU required.

Sheet Vision (Senior Design)

ElectronJS · ReactJS · AWS · Python · computer vision

Application that reads sheet music, plays it back, and listens to the user in real time, providing feedback to help learners draw parallels between notation and sound.

Run Samurai, Run!

P5.js · HTML/CSS · PHP · AJAX · game loop development

Top-down 2D infinite side-scrolling runner for mobile. Players fight through enemy mobs for upgrades and high scores, with single-player and head-to-head multiplayer modes.